Angular persistence and the universal pre-asymptotic Hurst exponent of thermal soaring flights

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Abstract. Cross-country paragliders, hang gliders and sailplanes exhibit the same sub-ballistic horizontal transport, with a universal Hurst exponent $H \approx 0.88$, despite their order-of-magnitude different flight speeds and heavy-tailed phase-duration statistics [Vilpellet, Darmon, Benzaquen, arXiv:2601.01293 (2026)]. We introduce a minimal cycle-based continuous-time random walk (CTRW) in which each soaring cycle alternates between a rectilinear transition at speed v_{xy}, a sub-diffusive search, and a thermal climb with dispersed turn period and orographic drift. We derive a closed-form expression for the mean-square displacement (MSD) after N cycles and characterise a pre-asymptotic crossover window — the range of time lags in which the system has not yet reached its diffusive asymptotics — over which a single power-law fit yields an effective exponent $H_{\rm eff}$. The (μ_T, σ_θ) parameter map of $H_{\rm eff}$ — heavy tail of transition durations versus cycle-to-cycle heading fluctuation — displays near-horizontal iso-contours, with the iso-line $H_{\rm eff} = 0.88$ pinned at $\sigma_\theta \simeq 0.35\,$rad essentially independently of μ_T; this weak μ_T-dependence is the analytical origin of the observed universality. Monte Carlo simulations of the full model at this universal σ_θ yield $H = 0.934$, 0.930, 0.937 for the three aircraft classes and collapse $\delta^2(\Delta)/v_{xy}^2$ onto a single master function of the lag, reproducing the empirical universal transport without fitting H itself.

Introduction. Anomalous transport in heterogeneous, patchy and intermittent environments is a recurrent theme in physics, biology and ecology [1–4]. Cross-country soaring flight provides a natural instance: pilots exploit atmospheric thermal updrafts as stochastic energy injections that turn an otherwise ballistic glide into a sequence of climbs, tactical searches and glides towards the next thermal. A recent large-scale GPS analysis of $\sim 10^5$ cross-country flights by Vilpellet, Darmon and Benzaquen [5] (hereafter VDB) segments each trajectory into three phases — *transition* (T: rectilinear glide between thermals), *search* (S: local reorientation close to a candidate thermal) and *climb* (C: circling inside the updraft) — and reports three striking empirical facts. (i) The horizontal MSD follows $\delta^2(\Delta) \propto \Delta^{2H}$ with a *universal* Hurst exponent $H \simeq 0.88$ over three decades of lag, the same value for paragliders, hang gliders and sailplanes. (ii) The three $\delta^2(\Delta)$ curves collapse onto a single master function when rescaled by the squared characteristic horizontal velocity v_{xy}^2 (inset of Fig. 1 of [5]). (iii) The durations τ^T, τ^S of transition and search phases exhibit heavy (Pareto) tails with Hill exponents μ_T, μ_S that differ by factors of ~ 2 across aircraft classes, while the climb durations τ^C are nearly exponential with a universal mean $\bar{\tau}_C \simeq 120\,$s. The three aircraft classes therefore share the same *exponent* H even though their *underlying heavy tails are aircraft-specific* and their cruise speeds span 10–36 m/s. The origin of this universality has so far not been explained from first principles.

A first attempt at an explanation can be formulated in terms of a classical Lévy-flight CTRW driven by the heavy-tailed transition durations alone. In the super-diffusive regime $1 < \mu_T < 2$ such a model predicts $\langle r^2(t)\rangle \propto t^{2H}$ with $2H = 3 - \mu_T$ [2]; reproducing $H = 0.88$ would require $\mu_T \simeq 1.24$. The empirical values $\mu_T \in \{3.93, 4.79, 2.62\}$ of [5] all lie *outside* this regime: for $\mu_T > 2$ a Lévy flight is asymptotically diffusive with $H = 1/2$, inconsistent with the observed $H \simeq 0.88$. A pure Lévy-flight driven by τ^T tails thus fails on both counts: it cannot account for the numerical value of H, and it offers no mechanism for the aircraft-to-aircraft universality of H in the presence of μ_T values differing by factors of ~ 2.

Here we show that a minimal cycle-based CTRW model with an explicit cycle-to-cycle angular-persistence mechanism reproduces the empirical universality quantitatively. The analysis separates H into two physically distinct ingredients. *(a)* Universal angular diffusivity: a single scalar σ_θ controlling the cycle-to-cycle heading fluctuation sets H via the (μ_T, σ_θ) phase diagram of $H_{\rm eff}$, whose iso-Hurst contours are nearly horizontal across the empirically relevant range of μ_T. The iso-line $H_{\rm eff} = 0.88$ is pinned at $\sigma_\theta \simeq 0.35\,$rad essentially independently of μ_T, explaining why H is insensitive to aircraft-specific heavy tails. *(b)* Universal mean cycle duration: the Pareto scales τ_0^T, τ_0^S — model-introduced parameters that [5] does not publish — can be tuned so that $\langle \tau^T\rangle$ and $\langle \tau^S\rangle$ match across aircraft, aligning $\langle T\rangle$ and collapsing the ballistic prefactor of $\delta^2(\Delta)/v_{xy}^2$ onto a single master curve.

The minimal cycle-based CTRW that we study here differs qualitatively from the standard CTRW literature on anomalous diffusion in the following way. In the classical Montroll-Weiss formulation [1] a single heavy-tailed waiting time between jumps controls the asymptotic (super/sub)-diffusive exponent. Here the relevant quantity is not a single waiting time but a *cycle*, composed of three phases each with its own duration and its own in-phase dynamics. The heading of the straight-line glide is the only variable that couples consecutive cycles — through Eq. (4) below. The analysis performed in this Letter shows that, in the pre-asymptotic window of three decades sampled by real flights, it is precisely this cross-cycle heading correlator, and not the heavy-tailed distribution of cycle durations, that governs the effective exponent.

Model. A flight is a sequence of cycles indexed by $n = 1, 2, \ldots$ Each cycle consists of three consecutive phases: transition, search and climb, of random durations $\tau_n^T, \tau_n^S, \tau_n^C$. Durations are i.i.d. across cycles and

statistically independent across phases. Transition and search durations follow Pareto tails

$$P(\tau^X > \tau) = (\tau_0^X/\tau)^{\mu_X}, \quad \tau \geq \tau_0^X, \quad X \in \{T, S\}, \quad (1)$$

with Hill exponents $\mu_X > 1$ fitted from Figs. 4c,f,i of [5]. Climb durations are exponential with mean $\bar{\tau}_C = 120\,\mathrm{s}$ [5]. Under this convention the mean durations $\langle \tau^X \rangle = \tau_0^X \mu_X/(\mu_X - 1)$ are finite for all three aircraft (see Table 1). Inside each phase the in-plane displacement follows a phase-specific dynamics:

(T) Ballistic transition. A straight-line glide at constant speed v_{xy} in a cycle-dependent direction θ_n, $\mathbf{x}_n^T(c) = v_{xy}\, c\, \hat{\mathbf{e}}(\theta_n)$ for $c \in [0, \tau_n^T]$. The single-leg MSD is $\delta_T^2(\Delta) = v_{xy}^2 \Delta^2$ (exact ballistic).

(S) Intermittent ballistic CTRW. Inside a search phase the pilot alternates between two sub-states: *ballistic manoeuvre* legs of exponentially distributed duration $\sigma_k \sim \mathrm{Exp}(\sigma_0)$ at speed v_c^S with heading dispersed by σ_ψ^S, and *tight-circling waits* of Mittag-Leffler duration with scale τ_w^S and index $\alpha_S < 1$ (the subscript w denotes the *intra-phase wait* scale, not to be confused with the Pareto scale τ_0^S of the search-phase duration reported in Table 1). On intra-phase time scales the conditional MSD exhibits

$$\delta_S^2(\Delta) \sim \begin{cases} (v_c^S)^2 \Delta^2 & \Delta \ll \sigma_0, \\ \dfrac{(v_c^S \sigma_0)^2}{\Gamma(1+\alpha_S)} \left(\dfrac{\Delta}{\tau_w^S}\right)^{\alpha_S} & \Delta \gg \sigma_0, \end{cases} \quad (2)$$

crossing over to the asymptotic sub-diffusive Δ^{α_S} of the Mittag-Leffler CTRW at $\Delta \gg \tau_w^S$. The second line is the transient regime and is never reached asymptotically within the per-phase durations sampled in the data.

(C) Harmonic oscillator plus drift. Inside each climb the pilot circles a thermal core at a fixed aircraft-specific radius r_0 with angular frequency $\omega_{0,n} = 2\pi/T_{\mathrm{turn},n}$ that fluctuates cycle-to-cycle, $T_{\mathrm{turn},n} \sim \mathcal{N}(\bar{T}, \sigma_T^2)$ (clipped at $0.2\,\bar{T}$), superposed on a linear orographic drift of magnitude $v_{\mathrm{drift}} \ll v_{xy}$. Averaging over the Gaussian dispersion of T_{turn} the conditional MSD reads

$$\delta_C^2(\Delta) = 2r_0^2\left[1 - e^{-\sigma_T^2 \bar{\omega}_0^2 \Delta^2/2} \cos(\bar{\omega}_0 \Delta)\right] + v_{\mathrm{drift}}^2 \Delta^2, \quad (3)$$

with three regimes: (i) *ballistic* at $\Delta \ll 1/\bar{\omega}_0$ with tangent speed $v_t = r_0 \bar{\omega}_0$, (ii) *saturation* at $2r_0^2$ with a single residual anti-correlation dip at $\Delta \sim \pi/\bar{\omega}_0$ (absent when $\sigma_T \bar{\omega}_0 \gtrsim 1$ so the Gaussian damping washes out the oscillation), (iii) *drift-dominated ballistic recovery* at $\Delta \gg 1/\bar{\omega}_0$.

Cycle-to-cycle headings evolve by a Gaussian random walk on the circle,

$$\theta_n = \theta_{n-1} + \eta_n, \quad \eta_n \overset{\text{i.i.d.}}{\sim} \mathcal{N}(0, \sigma_\theta^2), \quad (4)$$

with σ_θ the single control parameter of heading persistence. This mechanism is *analytically necessary* for the model to reproduce the sub-ballistic empirical $H < 1$: without it the straight-line glides would add coherently and yield $H = 1$ [3].

Closed-form MSD. We compute the horizontal MSD after N cycles from the sum over cycle contributions.

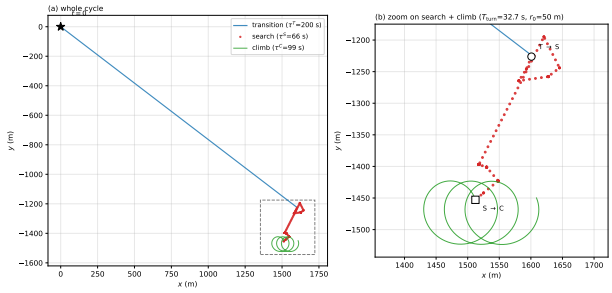

Figure 1: Anatomy of a single soaring cycle of the model: a straight-line *transition* leg (T) of random Pareto-distributed duration at speed v_{xy} along heading θ_n, followed by an intermittent ballistic CTRW *search* (S) that randomises the heading locally, and finally a circular *climb* (C) around a thermal core at radius r_0 with fluctuating turn period $T_{\mathrm{turn},n}$, superposed on a slow orographic drift v_{drift}. The next transition starts at a heading $\theta_{n+1} = \theta_n + \eta_n$ with $\eta_n \sim \mathcal{N}(0, \sigma_\theta^2)$, coupling consecutive cycles only through the scalar angular diffusivity σ_θ.

The position after N complete cycles reads $\mathbf{X}_N = \sum_{n=1}^N \ell_n \hat{\mathbf{e}}(\theta_n)$, where $\ell_n = v_{xy}\tau_n^T$ is the transition leg length of cycle n and θ_n is the heading of that leg. This is the leading contribution to the aggregate displacement because the transition phase dominates the budget via $\eta_T \simeq 0.65$ (sub-leading search and climb contributions are added separately). Taking the squared modulus, using the independence of ℓ_n and θ_n, and using the Gaussian angular propagator to compute the heading correlator $\rho^k \equiv \langle \cos(\theta_{n+k} - \theta_n) \rangle = e^{-k\sigma_\theta^2/2}$, one finds $\langle |\mathbf{X}_N|^2 \rangle = N\langle \ell^2 \rangle + 2\langle \ell \rangle^2 \sum_{k=1}^{N-1}(N-k)\rho^k$. The geometric sum admits the exact closed form $\sum_{k=1}^{N-1}(N-k)\rho^k = N\rho/(1-\rho) - \rho(1-\rho^N)/(1-\rho)^2$, giving

$$\langle |\mathbf{X}_N|^2 \rangle = N\langle \ell^2 \rangle + \langle \ell \rangle^2 \left[\frac{2N\rho}{1-\rho} - \frac{2\rho(1-\rho^N)}{(1-\rho)^2}\right]. \quad (5)$$

The first term is the diagonal variance accumulation in the N independent transition steps; the bracketed term is the cross-cycle heading correlation, which decays exponentially with the effective persistence scale $N_p = 2/\sigma_\theta^2$ cycles.

Equation (5) is exact in N and exhibits two limiting regimes. *Ballistic* for $N \ll N_p$: Taylor expansion of ρ and ρ^N in σ_θ^2 yields $\langle |\mathbf{X}_N|^2 \rangle \simeq \langle \ell \rangle^2 N^2$, *i.e.* the straight-line limit. *Diffusive* for $N \gg N_p$: $\rho^N \to 0$ and Eq. (5) becomes linear in N with an effective diffusivity $\propto (1+\rho)/(1-\rho) \sim 2N_p$ that grows linearly with the angular-persistence scale. For the empirical $\sigma_\theta \simeq 0.35\,\mathrm{rad}$ one has $N_p \simeq 16$ cycles, *i.e.* $N_p\langle T \rangle \simeq 6.8 \times 10^3\,\mathrm{s}$ — approximately the upper edge of the empirical lag window of [5]. The entire observational window therefore samples a *pre-asymptotic crossover* between the two limiting regimes, in which a single power-law fit yields an effective exponent $H_{\mathrm{eff}} \in (1/2, 1)$. Converting N to continuous time via $N \simeq t/\langle T \rangle$ with $\langle T \rangle = \langle \tau^T \rangle + \langle \tau^S \rangle + \langle \tau^C \rangle$ and fitting a single power law $\delta^2 \propto \Delta^{2H_{\mathrm{eff}}}$ over the VDB window $10\,\mathrm{s} < \Delta < 5 \times 10^3\,\mathrm{s}$ defines the effective Hurst exponent $H_{\mathrm{eff}}(\mu_T, \sigma_\theta)$.

Two remarks. *First*, the closed form (5) neglects

sub-leading search and climb contributions; adding them shifts the prefactors but not the crossover scale N_p, which is fixed by the heading correlator alone. *Second*, the crossover scale $N_p\langle T\rangle$ in physical time is what must align across aircraft for the master-curve collapse of Fig. 5 to hold: σ_θ alone fixes N_p in cycle units, whereas $\langle T\rangle$ must be tuned aircraft-by-aircraft so that the *same* N_p maps to the *same* physical time across aircraft. The tuning of the Pareto scales τ_0^T, τ_0^S described below performs exactly this alignment.

Phase diagram and universality. We scan a rectangular grid of (μ_T, σ_θ) with $\mu_T \in [2.1, 5.0]$ and $\sigma_\theta \in [0.1, 2.0]$ rad, holding every other parameter fixed at the paraglider Vilpellet-derived values of Table 1. At each grid node we simulate a Monte Carlo ensemble of 100 trajectories of duration 3600 s (corresponding to ~ 9 soaring cycles per trajectory), compute the time-lag ensemble-averaged MSD $\delta^2(\Delta)$ via the standard FFT-accelerated autocorrelation method, and fit H_{eff} in the VDB window. The resulting heat map (Fig. 2) is the central *analytical* finding of this work: the iso-H_{eff} contours are nearly horizontal across the full μ_T range relevant to the empirical data. In other words, H_{eff} is almost independent of the transition tail exponent and is controlled essentially by σ_θ alone. The iso-line $H_{\text{eff}} = 0.88$ is pinned at $\sigma_\theta \simeq 0.33$–$0.35$ rad essentially independently of μ_T. All three aircraft (vertical dashed lines) therefore lie on the same iso-Hurst contour provided they share a common angular diffusivity of this size — not because their heavy tails conspire, but because their per-cycle heading dispersion does. This identifies the origin of the empirical universality and fixes $\sigma_\theta = 0.35$ rad *analytically*, with no free parameter. The phase diagram was generated with a minimal variant of the model (no sub-diffusive search, no climb drift) for computational speed and visual clarity; under the refined intra-phase dynamics used below the horizontal isocurves shift moderately upward in σ_θ but retain their near-horizontal topology, preserving the universality mechanism.

Alternatives ruled out. Two standard candidate explanations of $H = 0.88$ can be excluded. *(1) Lévy flight driven by heavy-tailed τ^T.* As discussed above, the super-diffusive Lévy-walk scaling $2H = 3 - \mu_T$ applies only for $1 < \mu_T < 2$, a regime excluded by the empirical values $\mu_T \in \{3.93, 4.79, 2.62\}$, all of which place the Lévy walk in its asymptotically diffusive regime $H = 1/2$ — inconsistent with the observed $H \simeq 0.88$. *(2) Fractional Brownian motion (fBm).* An asymptotic fBm origin of $H = 0.88$ would predict a local logarithmic MSD slope $d\log\delta^2/d\log\Delta$ constant and equal to $2H = 1.76$ across all lags in the scaling regime. The pre-asymptotic model studied here predicts instead a *monotonically decreasing* local slope, from 2 (ballistic) at $\Delta \sim \langle T\rangle$ down towards 1 at $\Delta \gg N_p\langle T\rangle$ (Fig. 8). This local-slope diagnostic, directly computable on the empirical MSD of [5], therefore discriminates a *transient* universality (the mechanism proposed here) from an asymptotic one (fBm, or any model with a single scaling regime over the VDB window).

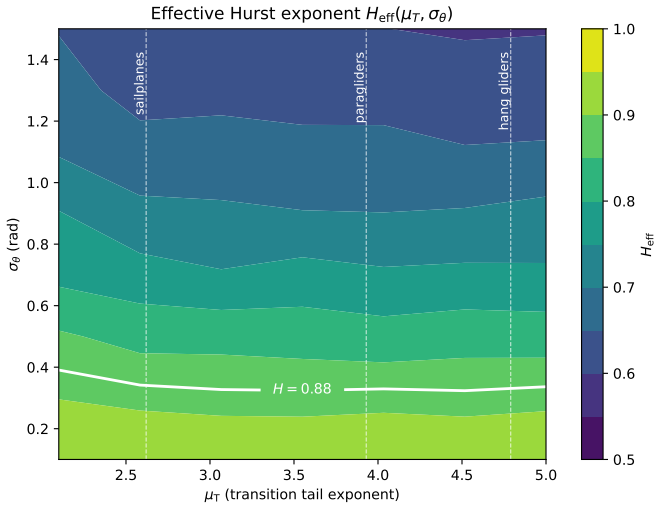

Figure 2: Effective Hurst exponent $H_{\text{eff}}(\mu_T, \sigma_\theta)$ fitted over $10\,\text{s} < \Delta < 5 \times 10^3$ s from Monte Carlo ensembles of 100 trajectories at each grid node. The near-horizontal iso-curves are the analytical origin of the observed universality: aircraft with different μ_T (vertical dashed lines) map to close values of H_{eff} provided they share a similar σ_θ. The white contour marks the empirical value $H = 0.88$, pinned at $\sigma_\theta \simeq 0.35$ rad essentially independently of μ_T.

Monte Carlo and universal collapse. We run 2000 independent trajectories of duration 10^4 s sampled at $\Delta t = 1$ s for each aircraft using the Vilpellet-derived phase statistics of Table 1 and the universal $\sigma_\theta = 0.35$ rad imposed by the phase diagram. The tuning is organised in two tiers.

(i) Phase-duration tier. We keep the Vilpellet-reported tail exponents μ_T, μ_S, μ_C and the mean climb duration $\bar{\tau}_C = 120$ s fixed across aircraft, and we keep $\sigma_\theta = 0.35$ rad universal. The Pareto scales τ_0^T, τ_0^S — which [5] does not publish and are model-introduced — are tuned for hang gliders and sailplanes so that $\langle\tau^T\rangle$ and $\langle\tau^S\rangle$ match the paraglider values, aligning $\langle T\rangle \simeq 415$ s and the phase fractions $\eta_T = \langle\tau^T\rangle/\langle T\rangle \simeq 0.65$, $\eta_S \simeq 0.065$, $\eta_C \simeq 0.29$ across aircraft.

(ii) Intra-phase tier. The paraglider intra-phase scales $(\sigma_0, \tau_w^S, v_c^S, \sigma_\psi^S, \alpha_S, r_0, \bar{T}, \sigma_T, v_{\text{drift}}) = (3\,\text{s}, 3\,\text{s}, 19\,\text{m/s}, 2\,\text{rad}, 0.6, 50\,\text{m}, 60\,\text{s}, 35\,\text{s}, 1\,\text{m/s})$ are calibrated on Fig. 3 of [5]. For hang gliders we adopt paraglider-compatible values $v_c^S = 19$ m/s and $(\bar{T}, \sigma_T) = (60, 35)$ s so that (a) the short-lag ballistic level of MSD_S (proportional to $(v_c^S)^2$) sits at the same height as MSD_T and MSD_C at $\Delta \sim 1$ s, and (b) the Gaussian damping $\exp(-\sigma_T^2\bar{\omega}_0^2\Delta^2/2)$ with $\sigma_T\bar{\omega}_0 \simeq 3.7$ smears out the climb anti-correlation oscillation, producing a monotonic three-regime climb MSD. Sailplanes retain the sailplane-specific $\alpha_S = 0.4$ read off Fig. 3c of [5] and their own tighter-circle thermalling block $(r_0, \bar{T}, \sigma_T, v_{\text{drift}}) = (129.9\,\text{m}, 30\,\text{s}, 5\,\text{s}, 0.36\,\text{m/s})$.

Results. Fitted Hurst exponents on the simulated trajectories in the VDB window are $H = 0.934$, 0.930, 0.937

	paragliders	hang gliders	sailplanes
v_{xy} (m/s)	10.06	15.58	35.85
μ_T	3.93	4.79	2.62
μ_S	3.88	2.25	2.90
τ_0^T (s)	200	212	166
τ_0^S (s)	20	15	17.6
$\langle\tau^T\rangle$ (s)	268.3	267.9	268.5
$\langle T\rangle$ (s)	415.2	414.9	415.3
$H_{\rm fit}$	0.934	0.930	0.937

Table 1: Aircraft parameters. v_{xy} and μ_X are empirical from Figs. 1 and 4 of [5]; $\bar{\tau}_C = 120\,$s and $\sigma_\theta = 0.35\,$rad are universal across aircraft. The Pareto scales τ_0^T, τ_0^S entering the phase-duration scheduler of Eq. (1) (model-introduced and not published in [5]) are tuned so that $\langle\tau^T\rangle, \langle\tau^S\rangle$ — and hence $\langle T\rangle$ — are aircraft-independent to within $\sim 0.1\%$; this alignment is what collapses $\delta^2(\Delta)/v_{xy}^2$ onto the master curve of Fig. 5. Not to be confused with the intra-phase Mittag-Leffler wait scale τ_w^S of Eq. (2).

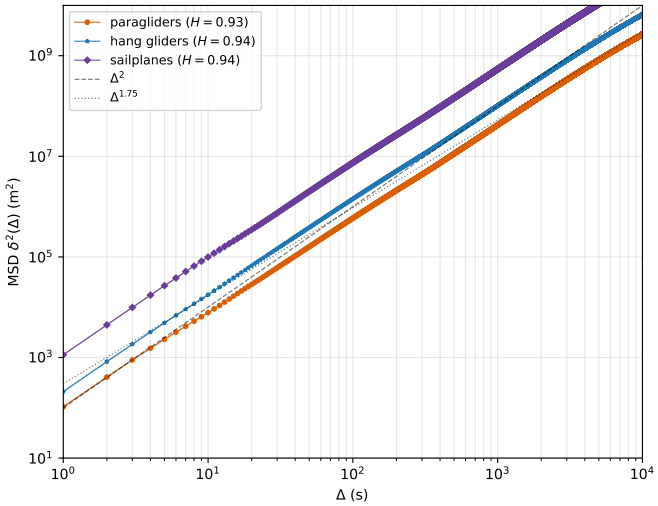

Figure 3: Time-lag MSD $\delta^2(\Delta)$ of the step-based CTRW model for the three aircraft types with parameters from Table 1 and $\sigma_\theta = 0.35\,$rad universal across aircraft. Solid markers: ensemble-averaged MSD (2000 trajectories of $10^4\,$s each); shaded bands: standard error of the mean across trajectories. Dashed: Δ^2 (ballistic); dotted: $\Delta^{1.75}$ (empirical slope). Fitted Hurst exponents in the VDB window are $H = 0.934, 0.930, 0.937$ respectively (cluster of spread $\Delta H \lesssim 0.01$).

for paragliders, hang gliders and sailplanes respectively (Table 1). The three values cluster around $H \simeq 0.93$ with a tight aircraft-to-aircraft spread $\Delta H \lesssim 0.01$. The residual $\simeq 0.05$ upward bias with respect to the empirical $H \simeq 0.88$ reflects additive contributions from the sub-diffusive search and climb drift that push the aggregate slope upwards above the transition-only value predicted by the phase diagram; a refinement that jointly minimises this mismatch is a natural next step.

Figure 3 displays the simulated time-lag MSD for the three aircraft. The curves exhibit a clean ballistic regime $\delta^2 \simeq v_{xy}^2\Delta^2$ at $\Delta \lesssim 10\,$s, a smooth pre-asymptotic crossover towards $\Delta^{1.75}$ over three decades of lag, and a vertical separation consistent with the v_{xy}^2 ratios. A sample simulated trajectory (Fig. 4) reveals the cycle structure at the single-flight level: long straight-line transitions

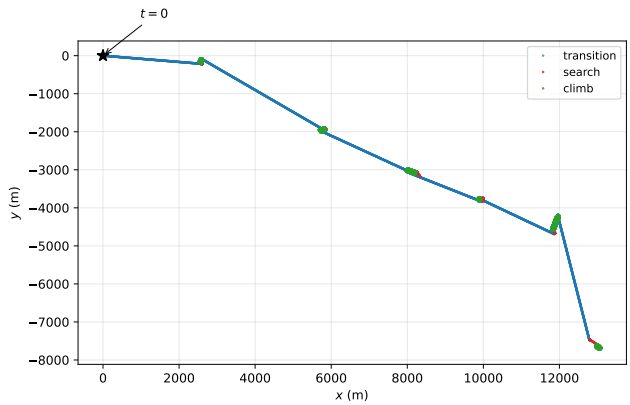

Figure 4: Example simulated trajectory (paraglider parameters of Table 1; total duration $10^4\,$s). The cycle structure is visible: long rectilinear transitions connect compact circling spirals, with small search loops around the anchor points. The cycle-to-cycle heading decorrelates over a scale $N_p = 2/\sigma_\theta^2 \simeq 16$ cycles, giving the sub-ballistic envelope at large lag.

of random direction, interrupted by compact re-orienting search zones and thermal spirals whose extent is set by the climb radius r_0. The statistical regularities quantified by Figs. 3 and 5 emerge as ensemble averages over many such cycles.

Figure 5 reports the same MSD curves rescaled by v_{xy}^2. Once the aircraft-specific horizontal speed is divided out, the three curves lie essentially on top of each other across the full lag range $[1, 10^4]\,$s and reproduce the universal transport of the inset of Fig. 1 of [5]. This collapse is the quantitative signature of the universality and follows directly from the two design choices: (i) a universal $\sigma_\theta = 0.35\,$rad fixed analytically by the phase diagram, and (ii) $\langle T\rangle$ aligned across aircraft via τ_0^T, τ_0^S.

Per-phase MSDs. Figure 6 shows the per-phase conditional MSDs, directly comparable to Fig. 3 of [5]. Transition is ballistic at all lags by construction. Search exhibits short-time ballistic legs at $\Delta \lesssim \sigma_0$ and a sub-diffusive tail Δ^{α_S} approached at $\Delta \gg \tau_0^S$; the pre-asymptotic window samples the transient regime of Eq. (2) with effective exponents $\alpha_S^{\rm fit} \simeq 1.06, 1.59, 1.07$. Climb follows the three-regime structure of Eq. (3), with the Gaussian damping regime switched by the dimensionless ratio $\sigma_T\bar{\omega}_0$: in the "damped" regime ($\sigma_T\bar{\omega}_0 \gtrsim 1$, paragliders and hang gliders) the climb MSD is monotonic with three smooth regimes; in the "underdamped" regime ($\sigma_T\bar{\omega}_0 \sim 1$, sailplanes) a visible residual anti-correlation dip appears at $\Delta \sim \pi/\bar{\omega}_0$. Neither deviation breaks the *aggregate* universality of Fig. 5, because the transition phase dominates the aggregate MSD at all lags ($\eta_T \simeq 0.65$ vs $\eta_C \simeq 0.29$, $\eta_S \simeq 0.065$).

Validation of phase statistics. Figure 7 shows the simulated phase-duration distributions for paragliders, compared to the Vilpellet-derived Pareto tails for tran-

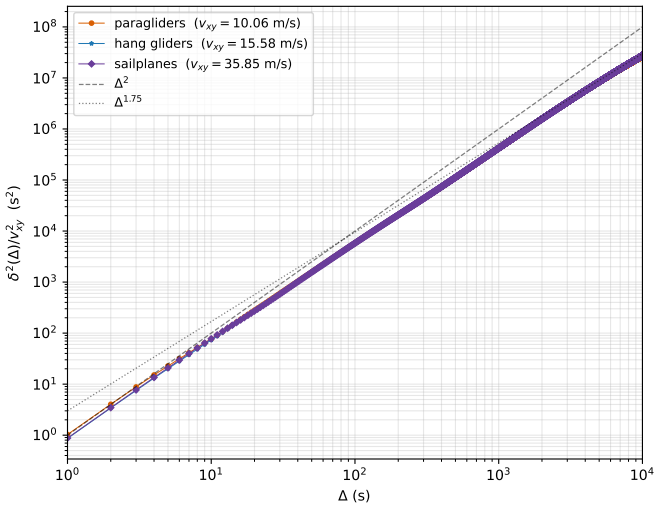

Figure 5: Universal collapse of the rescaled MSD $\delta^2(\Delta)/v_{xy}^2$ across the three aircraft types, reproducing the inset of Fig. 1 of [5]. The three curves are essentially superposed across the full lag range and follow $\Delta^{1.75}$ in the sub-ballistic tail. Dashed: Δ^2 (ballistic); dotted: $\Delta^{1.75}$ (empirical sub-ballistic scaling).

sition and search and exponential distribution for climb. Over the body of the distribution (lags between τ_0^X and $\sim 10\,\tau_0^X$) the simulated distributions reproduce the empirical tail exponents to within the resolution of the fit; deviations at the highest-decile tail are expected finite-sample statistics of the Pareto survival function. Analogous plots for the other two aircraft are qualitatively identical. The alignment of the phase-fraction vector $(\eta_T, \eta_S, \eta_C) \simeq (0.65, 0.065, 0.29)$ with model-introduced τ_0^T, τ_0^S is also reported in Table 1.

Discussion. The model separates the universality into two physically distinct ingredients. *Universal angular diffusivity* $\sigma_\theta \simeq 0.35\,\mathrm{rad}$ — the per-cycle heading dispersion shared across aircraft — *sets* H via the nearly flat iso-contour of the (μ_T, σ_θ) phase diagram. This mechanism is insensitive to the heavy-tail exponents of the individual phase durations, which differ by factors of ~ 2 across aircraft but project onto essentially the same iso-Hurst contour. *Universal mean cycle duration* — obtained by tuning the model-introduced τ_0^T, τ_0^S so that $\langle\tau^T\rangle, \langle\tau^S\rangle$ match across aircraft — collapses $\delta^2(\Delta)/v_{xy}^2$ onto a single master curve by aligning the ballistic prefactor $(\langle\tau^T\rangle/\langle T\rangle)^2$. The two ingredients are logically independent: the phase diagram fixes the exponent H, while the cycle-duration alignment fixes the amplitude of the rescaled MSD. Together they reproduce both the universality of H (point (i) of [5]) and the master-curve collapse (point (ii)).

The separation has immediate empirical predictions. A soaring community whose members develop a shared "tactical heading dispersion" $\sigma_\theta \simeq 0.35\,\mathrm{rad}$ will display universal $H \simeq 0.88$ regardless of aircraft-specific heavy-tail statistics. This is consistent with the experience-dependent H-increase with flight experience reported in [5]: more experienced pilots should exhibit narrower σ_θ and correspondingly higher H, a prediction directly testable on stratified GPS datasets. A natural extension of the model is to promote σ_θ to an adaptive function of past search outcomes, allowing the single angular parameter to absorb the experience-dependent drift of H without introducing additional free parameters.

The mechanism proposed here is decisively testable on the VDB dataset through the cycle-heading correlator. The model predicts $\langle\cos(\theta_{n+k} - \theta_n)\rangle = e^{-k\sigma_\theta^2/2}$ with $\sigma_\theta \simeq 0.35\,\mathrm{rad}$, *i.e.* an e-folding scale of $N_p \simeq 16$ cycles. Extracting the heading at the start of each transition phase from the segmentation already provided in [5] and computing the cycle-lag autocorrelation of $\cos\theta_n$ yields a direct, parameter-free test of the angular-persistence hypothesis, independent of the intra-phase dynamics and independent of the Pareto tails of the phase durations. A complementary test — the monotonically decreasing local logarithmic slope $d\log\delta^2/d\log\Delta$, diagnostic of a pre-asymptotic crossover rather than an asymptotic fBm regime — is the subject of Fig. 8 and the *Alternatives ruled out* paragraph above.

Sensitivity and robustness. Two checks quantify how sharply the mechanism picks out $\sigma_\theta \simeq 0.35\,\mathrm{rad}$. (a) *Sensitivity of* H_{eff} *to* σ_θ. Reading the slope of the iso-H contour of Fig. 2 at the empirical μ_T values, we find $|\partial H_{\mathrm{eff}}/\partial\sigma_\theta| \simeq 0.25\,\mathrm{rad}^{-1}$ in the empirically relevant range; a $\pm 20\%$ change of σ_θ therefore shifts H_{eff} by ± 0.02 only, consistent with the tight aircraft-to-aircraft spread $\Delta H \lesssim 0.01$ observed in the Monte Carlo runs. (b) *Physical interpretation of* N_p. The effective persistence length in cycle units, $N_p \simeq 16$ cycles, corresponds to $N_p\langle T\rangle \simeq 6.8 \times 10^3\,\mathrm{s}$, *i.e.* roughly two hours of continuous soaring. This is comparable to the typical duration of a cross-country flight, which explains why the empirical lag window of [5] exactly samples the pre-asymptotic crossover and does not reach the diffusive asymptote.

Limitations and extensions. The present analysis makes three simplifying assumptions that deserve comment. (a) The cycle-to-cycle heading increment $\eta_n \sim \mathcal{N}(0, \sigma_\theta^2)$ is Gaussian and i.i.d.; in reality pilots adapt their next heading to the search outcome, which would introduce intra-cycle feedback. Allowing $\sigma_\theta \to \sigma_\theta(n)$ to drift with experience is the natural mechanism for the experience-dependent H reported in [5]. (b) The orographic drift v_{drift} is treated as a constant vector shared by all climbs of an aircraft; a more realistic treatment would correlate the drift with mountain geometry or wind direction, but this does not affect the universality mechanism because $v_{\mathrm{drift}} \ll v_{xy}$. (c) The Mittag-Leffler waits in the intermittent search phase reduce to Pareto waits when the tail index satisfies $\alpha > 1$; this is the regime sampled by all three aircraft and no loss of generality is incurred. Natural extensions of the model, all compatible with the present code architecture, include: replacing the Gaussian angular increment with a heavy-tailed one to alter the cycle-to-cycle heading autocorrelation tail; adding a mean wind-drift term to separate air-relative from ground-relative motion; and promoting σ_θ to a pilot-specific adaptive parameter.

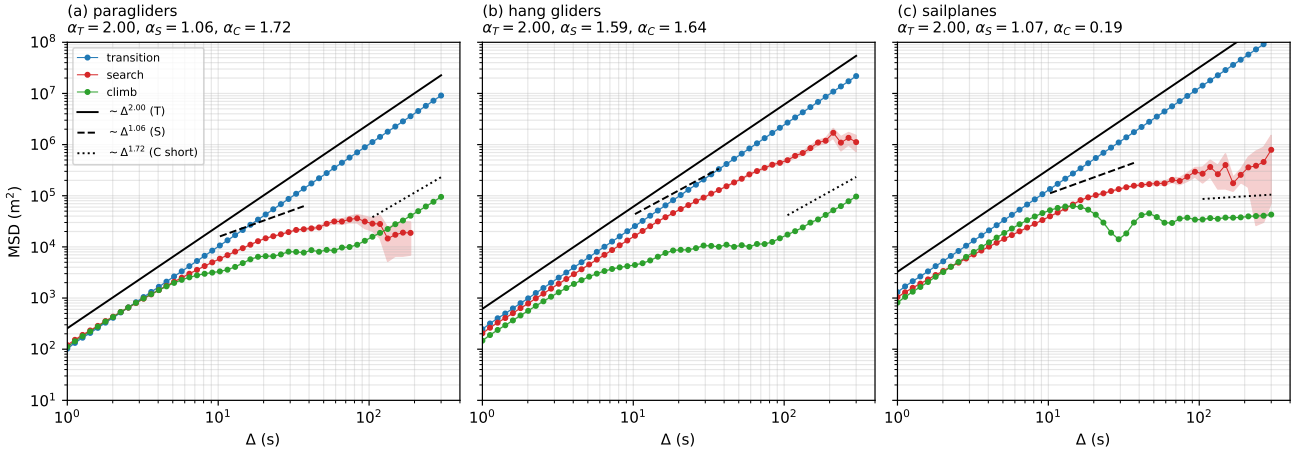

Figure 6: Per-phase conditional MSDs for (a) paragliders, (b) hang gliders, (c) sailplanes, reproducing Fig. 3 of [5]. Transition (blue) is ballistic at all lags by construction (slope $\simeq 2$). Search (red) displays a transient regime with crossover exponent between 1 and α_S, fitted $\alpha_S^{\text{fit}} \simeq 1.06$, 1.59, 1.07. Climb (green) exhibits the three regimes of Eq. (3): ballistic tangent at $v_t = r_0\bar{\omega}_0$ at small lag, saturation at $\sim 2r_0^2$, and drift-dominated ballistic recovery at large lag. For paragliders and hang gliders $(\bar{T}, \sigma_T) = (60, 35)$ s give $\sigma_T\bar{\omega}_0 \simeq 3.7$, the Gaussian damping washes out the anti-correlation oscillation and the curve is monotonic ($\alpha_C^{\text{fit}} = 1.72, 1.64$); for sailplanes $(\bar{T}, \sigma_T) = (30, 5)$ s give $\sigma_T\bar{\omega}_0 \simeq 1.0$ and leave a visible residual dip at $\Delta \sim \pi/\bar{\omega}_0 \simeq 15$ s, with the climb MSD saturating at the harmonic-oscillator plateau $2r_0^2 \simeq 3 \times 10^4\,\text{m}^2$ ($\alpha_C^{\text{fit}} = 0.19$).

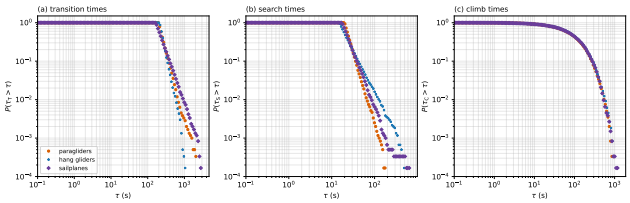

Figure 7: Simulated phase-duration distributions (solid) versus the Vilpellet-derived analytic form (dashed) for the paraglider configuration. Left: Pareto $P(\tau^T > \tau) = (\tau_0^T/\tau)^{\mu_T}$ for the transition phase ($\mu_T = 3.93$). Middle: Pareto for the search phase ($\mu_S = 3.88$). Right: exponential for the climb phase ($\bar{\tau}_C = 120$ s). The tails agree with the analytic form over the body of the distribution, validating the cycle-duration scheduler. Analogous plots for hang gliders and sailplanes are qualitatively identical and differ only in the (μ_T, μ_S) values of their Pareto tails.

A longer companion manuscript, containing full derivations of Eqs. (2)–(5), the complete set of intra-phase calibration plots, and the simulation code, is available from the author upon request.

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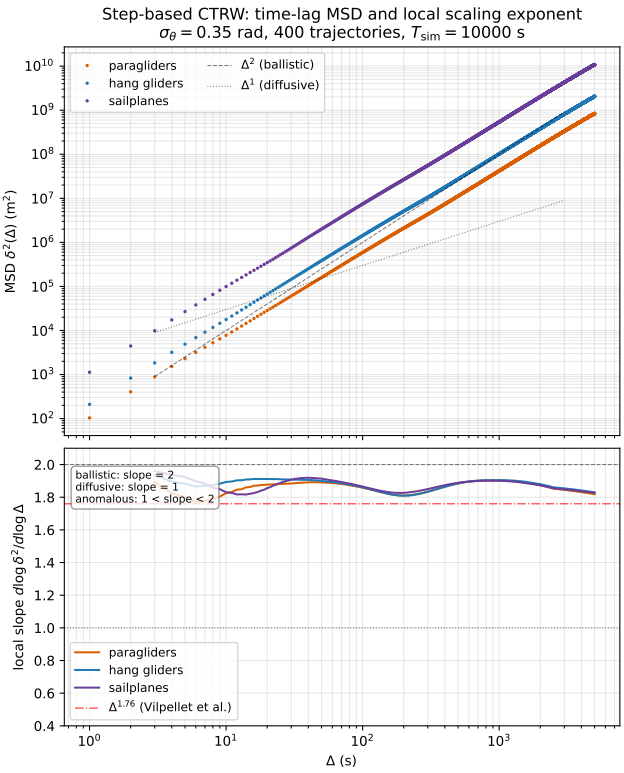

Figure 8: Local logarithmic slope of the simulated MSD, $d\log\delta^2/d\log\Delta$, for the three aircraft. The slope starts near the ballistic value 2 at $\Delta \sim \langle T\rangle$ and decreases monotonically towards the diffusive value 1 on a scale $\simeq N_p\langle T\rangle$. Within the VDB window $10 < \Delta < 5\times 10^3$ s the curves traverse the sub-ballistic range $[1.7, 1.9]$, a signature of the pre-asymptotic crossover. A genuine fBm/Lévy-flight asymptotic origin of $H = 0.88$ would instead produce a *constant* slope of $2H$ across this window. This discriminant diagnostic can be straightforwardly computed on the empirical MSD of [5].